Predicting thermal resistance of solder joints based on Scanning Acoustic Microscopy using Artificial Neural Networks

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***Abstract*—Scanning Acoustic Microscopy (SAM) measurements of thermally aged LED solder joints are translated to the thermal properties of the sample as characterized by Transient Thermal Analyses (TTA) using Artificial Neural Networks in order to improve the comparability of these two measurement methods. The dataset of 1800 samples with five solder pastes and nine LED types is used to study the inter- and extrapolation abilities of the trained models with respect to differences in solders and component structure. The effect of solder joint degradation due to thermal shock cycles on the ability of the model to translate is also studied with four different aging states. The architecture used is a combination of convolutional layers with max pooling and fully connected layers.**

Keywords— Solder Joints, LED, non-destructive testing, Machine Learning, Scanning Acoustic Microscopy (SAM), Transient Thermal Analysis (TTA), Convolutional Neural Network (CNN)

I. Introduction

Reliability is a key concern for modern electronics, and to classify the state of the component before total failure, good measurement methods are critical. Non-destructive measurements are especially useful, since they allow checking for production errors and monitoring over the aging process to better understand degradation. For LEDs, the most critical section for reliability is generally the solder joint. One well suited measurement approach is Scanning Acoustic Microscopy (SAM) [1], which produces a 3D or 2D image that can be easily interpreted by humans to estimate the state of the joint without damaging the component. However, the process is complex and slow, since it requires immersion in a water bath and is also prone to interference from the board or the measured component. A faster, less work-intensive measurement is the Transient Thermal Analysis (TTA). However, TTA is much harder to interpret and use as a representation of the state of the solder joint than the direct visualisation of a SAM image.

In this work, we use artificial neural networks to make a bridge between these two measurements, so that SAM images can be interpreted in terms of a TTA result. The broader goal is to create a two-way connection between the two measurement approaches, so future work will consider translating TTA data to SAM derived values, for example the ratio of crack area to intact solder pad.

This paper focuses on evaluating the ability of the model to inter- and extrapolate from the dataset, since this is a critical necessity for broader use. Specifically, the interpolation between solders, between different LED types, and between different to aging times is considered.

The section II explains how the data is collected, both in regard to the measurement methods of SAM and TTA, and in the design of the study. Section III gives an overview of the theoretical background of the neural networks used here and discusses some other works that look at SAM data using neural networks. Section IV discusses the architecture used in this paper and the details of how the network was trained. Finally, section V discusses the results for the extrapolation based on the solder, LED type and cycle count.

II. Data Collection

The following section briefly describes the data collection process. For that, first the measurement techniques used to collect the data used in this paper are discussed, and then the electronic components and the study design is described. For a more in-depth description of the electronic components, soldering, and initial characterisation see [2].

A. Scanning Acoustic Microscopy

Scanning Acoustic Microscopy is a non-destructive imaging method that allows inspection of the internal structure of components using focused sound waves (50 MHz for this dataset), similar to ultrasound used in medicine. Parts of the acoustic wave send by the emitter are reflected at any interface between two different materials according to the change in acoustic impedance. Based on the timing and strength of the reflections, the composition of the specimen at a single point is computed. The complete information is collected by scanning across in x and y direction [3] [4]. In this paper, we use 2D images at the depth of the solder joint.

SAM can show voids, non-wetted areas and cracks, since the interface to the gas/water filling the crack/void produces a reflection. However, resolution is worse than X-Ray, and depends on the frequency. This is a trade-off, since higher frequency allows better resolution but lower penetration.

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The specimen is placed in water to help transmission from emitter to specimen and back, so after measurement the part has to be dried. This and the time required for the scanning means that a SAM image requires a significant amount of time and work investment.

Examples of SAM images for all LED types in the study are shown in Table 2. The shown images are cropped to only the region of the LED itself. The original images contain a larger part of the surrounding, but those are irrelevant for the reliability of the joint. For our application, they are also cropped to save computational effort during the training of the network. The cropping is by LED type, which means the individual LED position changes slightly within the image due to manufacturing differences and slight changes in how the LED was positioned in the apparatus during the measurement.

Some of the measurements show artifacts due to scratches on the back of the printed circuit board (PCB) from the automatic height adjustment. This causes issues with the trigger and so at these positions the correct depth cannot the identified.

B. Transient Thermal Analysis

A quicker and easier way to characterize solder joints is to utilize the heat flow through the joint using Transient Thermal Analysis. The measurement is non-destructive, can be done in a few second, and only requires a voltage measurement. Analysis of the raw data is also significantly more straightforward. LEDs in particular are well suited to this measurement method, as they can serve as the necessary heat source; simplifying the measurement setup. This method has been used been used for measuring joint integrity during accelerated aging in other works [5] [6].

TTA measures the time-dependent thermal impedance Z_{th}, i.e. how easy heat can flow from the source to the sink. Defects such as cracks in the solder impede the flow by reducing the contact area and so can be detected. The further along the path of heat flow a defect is from the heat source, the later its effects will show in the measurement, which allows locating the defect to a position on the thermal path [7] [8].

To perform the measurement, the temperature over time $\mathrm{T_J(t)}$ at a junction is determined via the voltage $\mathrm{V_f(t)}$ at a fixed current using the temperature sensitivity SEN

$$\mathrm{T_J(t)} = \frac{\mathrm{V_f(t)}}{\mathrm{SEN}}.$$

$\mathrm{Z_{th}(t)}$ can then be calculated from the change in temperature since the start of the measurement

$$\Delta \mathrm{T_J(t)} = T_J(\mathrm{t}) - T_J\,(t_0)$$

using the change in heating power ΔP_{th}

$$\mathrm{Z_{th}(t)} = \frac{\Delta \mathrm{T_J(t)}}{\Delta \mathrm{P_{th}}}.$$

In our measurement setup, one TTA measurement consists of a heating phase and a measurement phase, visualized in Figure 1. We measure the cooling, since this requires only a small current and is therefore less noisy.

The final state value of the thermal impedance is the thermal resistance, which can also be computed from the temperature difference $\Delta \mathrm{T_J}$ between the steady-state values for the heating and measurement phases

$$R_{th} = Z_{th}(t \to \infty) = \frac{\Delta \mathrm{T_J}}{\Delta P_{th}}.$$

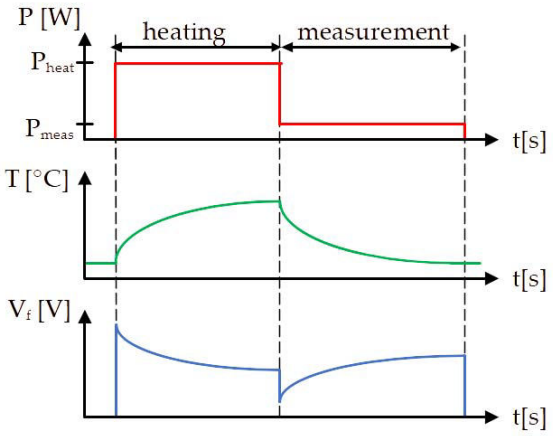


Figure 1: Behaviour of important quantities (electric power, temperature, measured voltage) across one full measurement.

The standard for TTA is described in JESD51-14 [9] and the TTA tester used in [10].

A useful way to further analyze the data is to consider the derivative with respect to the logarithmic time $z = \log(t)$. One reason to consider a logarithmic timescale is that the cooling can be approximated as an exponential decay, since the heat flow is proportional to the temperature difference to the environment.

We define the logarithmic derivative as

$$\mathrm{b}(z) = \frac{d}{dz} Z_{th}(z) = \frac{d}{dz} \frac{\Delta V_f(z)}{SEN * \Delta P_{th}}.$$

Using $B = \log(b)$ also makes interpretation easier, as changes can be interpreted in terms of R_{th}. There are also other reasons related to the measurement, for details see [7].

In practice, some further filter steps are needed. To compute Z_{th} the data is first smoothed with a running mean filter (window length 15) and the early time range of 0 to 20µs is extrapolated using a $\sqrt{t}$ -fit in the region of 20µs to 40µs to deal with the interference of switching from heating to measuring currents [9].

To compute the derivative b, Z_{th} is interpolated with cubic splines to 256 grid points uniformly distributed in z-space in the range between ln(2µs) and ln(1.95s) and then filtered with a Tikhonov regularization filter [11] with $\lambda = 0.00001$.

In addition to R_{th}, our model will predict B_{max}, the maximum value of $B(z)$, and z_{max}, the value of z for the peak value B_{max}.

C. Study Setup

For the underlying study, nine different LED types were soldered with five different solders (SAC105, SAC+BiIn, SAC305, SAC+Sb, SAC+SbBiNi, see Table 1) to Aluminum insulated metal substrate printed circuit and aged with thermal shock cycles. Each solder/LED type combination contributes 40 LEDs, for a total of 1800 LEDs.

The LEDs used were four Flip Chip with Solder Pads (FC-SP), three Flip Chip with Gold Bumps (FC-GB) and two Vertical Thin Film (VTF) LEDs. A listing of the LED properties is shown in Table 2, taken in a modified version

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from the initial paper describing the study [2]. That paper also contains a more detailed description of the structure of the LEDs, the data collection methodology, and the study as a whole, which for brevity we do not discuss in depth here.

The most critical factor is the carrier for actual LED die. The LEDs with a copper-based lead frame (FC-SP1 and VTF1) match the thermal expansion of the aluminum used as the main PCB material much more closely. The main driver of mechanical stress in the solder joint is the mismatch in the coefficients of thermal expansion of LED and PCB, which is significantly reduced for the copper carrier LEDs. This mechanical stress is in turn the main cause of crack propagation, and so these LEDs are also much more reliable compared to the LEDs using an AlN ceramic submount.

A second important factor is the metallization of the solder pad. All LEDs except for FC-GB2 and FC-GB3 have a thick metallization layer of about 80µm, while FC-GB2 and FC-GB3 use a copper thinfilm metallization. In the study, they showed a greater rate of damage accumulation.

The LEDs were subjected to shock cycles in an air-to-air chamber according to Profile G [12]of JEDEC standard JESD22-A104D, which means a temperature range from -40°C to 125°C and dwell times of 30 minutes at the high and low temperatures. The failure mode of the LEDs under the aging conditions were cracks in the solder. These can be seen directly in the SAM images, since the crack reflects the acoustic waves differently. They also influence the Z_{th} curve, since the reduced contact area due to the crack also reduces the heat flow.

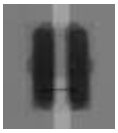
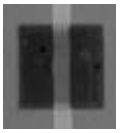
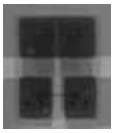
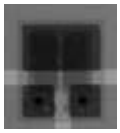
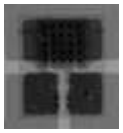
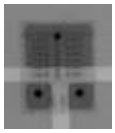
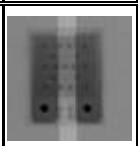
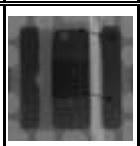
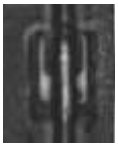
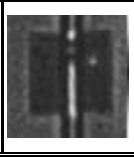
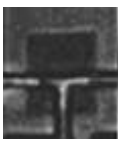
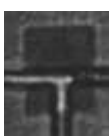
Measurements for both TTA and SAM are available at 0, 100, 500, 1000 and 1500 shock cycles, though this paper only uses data up to 1000 cycles as the last measurements were not yet taken.

In total, 7110 measurements were used for this study. The difference to the base sample size of 7200 (1800 LEDs measured at four cycles) is because of 90 LEDs (two LEDs

TABLE 1: SOLDER COMPOSITIONS

Solder	Additive material					
	Ag	Cu	Sb	Bi	Ni	In
SAC105	1.0	0.5	-	-	-	-
SAC107+BiNi	1.0	0.7	-	1.6	-	0.2
SAC305	3.0	0.5	-	-	-	-
SAC327+Sb	3.2	0.7	5.5	-	-	-
SAC387+SbBiNi	3.8	0.7	1.5	3.0	0.15	-

TABLE 2: LED PROPERTIES

Name in this paper		FC-SP1	FC-SP2	FC-SP3	FC-SP4	FC-GB1	FC-GB2	FC-GB3	VTF1	VTF2
Classification		Automotive	General	Automotive	Automotive	Automotive	Automotive	Automotive	Automotive	Automotive
Image										
Cross-Section										
Carrier		Copper	Ceramic	Ceramic	Ceramic	Ceramic	Ceramic	Ceramic	Copper	Ceramic
Pad Count		2	2	3	3	3	3	2	3	3
Thin Film Metallization		No	No	No	No	No	Yes	Yes	No	No
Size	[mm²]	1.92	2.56	5.0	3.03	4.37	2.61	2.61	14.06	2.85
Pad-Size	[mm²]	0.81	1.80	2.79	1.86	2.75	1.45	1.49	5.40	1.74
I_{Max}	[A]	0.7	2.0	1.5	1.5	1.5	1.2	1.2	1.5	1.5
V_f	[V]	2.75 ... 3.00	2.9	2.90 ... 3.50	2.8 ... 3.4	2.79 ... 3.03	2.9 ... 3.5	2.9 ... 3.9	3.00 ... 3.25	3.00 ... 3.25
$R_{th\text{-}JC\text{-}el.}$ (4)	[K/W]	4.75	6.0*	3.0	3.5	4.2	3.3	3.0	3.0	3.3
X-ray (not scaled)										
SAM (not scaled)										

* Value from the datasheet doesn't correspond to measurement results with approx. 2 K/W and also to the LED design with a ceramic material (AlN) like the others. It is assumed that the value is in the datasheet is erroneous. However, the supplier didn't respond after contact and reporting the error.

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per Solder/Component Type) removed at 500 cycles for cross sections that therefore couldn't be measured at 1000 cycles.

III. Neural Network Background

A neural network is a mathematical model to map a set of inputs to outputs built up from a number of layers, each performing a distinct mathematical operation [13]. Each layer has a set of parameters which are tuned by training/fitting the model on a set of training data to optimize performance of the overall network.

A. Fundamentals

The most basic type of neural network in common use is the Fully Connected layer. It corresponds to a matrix multiplication followed by the application of a nonlinear activation function. More precisely, for an input x, either the input of the model or the result of the previous layer, the output a is $a = \sigma(\mathrm{w} * \mathrm{x} + \mathrm{d})$, with the weights matrix w, bias vector d and activation function σ. w and d contain the trainable parameters. The number of trainable parameters of one layer therefore depends on the size of the input and the output.

The nonlinearity of σ is critical for the expressiveness of the model, as otherwise it would simply be a very complicated way of defining a polynomial model. This layer type is called 'fully connected' since every output is in principle computed based on all inputs, though training can bring a weight to or near 0 and eliminate a connection.

Convolutional Neural Networks (CNN) are often used to work with images, like the SAM data here. As the name says, it performs a convolution of the input with a kernel, which contains the trainable parameters. Typically, one layer contains several different kernels, which get combined into one output image (similar to how an RGB color image consists of three images that each show a single color). Each kernel detects one type of feature, for example vertical edges.

The advantage of CNN layers is that they automatically consider spatial relationships in the data by the nature of how a convolution works (only the points covered by the kernel are used to compute the new pixel), and that they have orders of magnitude fewer training parameters (one set of parameters applies to all inputs and does not scale with input size). The lower number of parameters makes training faster and less prone to overfitting. Since the output of a CNN is another image, the computational process is also more easily understood, which is difficult for typical NN models.

Often, CNNs are combined with Pooling layers. These combine several pixels into one and help shrink down the output to further reduce the number of parameters needed. Here, we use Max Pooling, which takes the maximum value in a box as the output. A 2D Max Pooling layer with side length 2 will therefore reduce the image dimension by half, quarters the number of pixels. They also give some resistance to small translational changes of the target objects, since several nearby pixels are grouped together and only the largest value is picked.

To evaluate the performance of the network, the output is compared to the true values using an error or loss function. Based on this, the trainable parameters are optimized using an optimization algorithm. This is repeated until the parameter converge when the (locally) best model parameters are found.

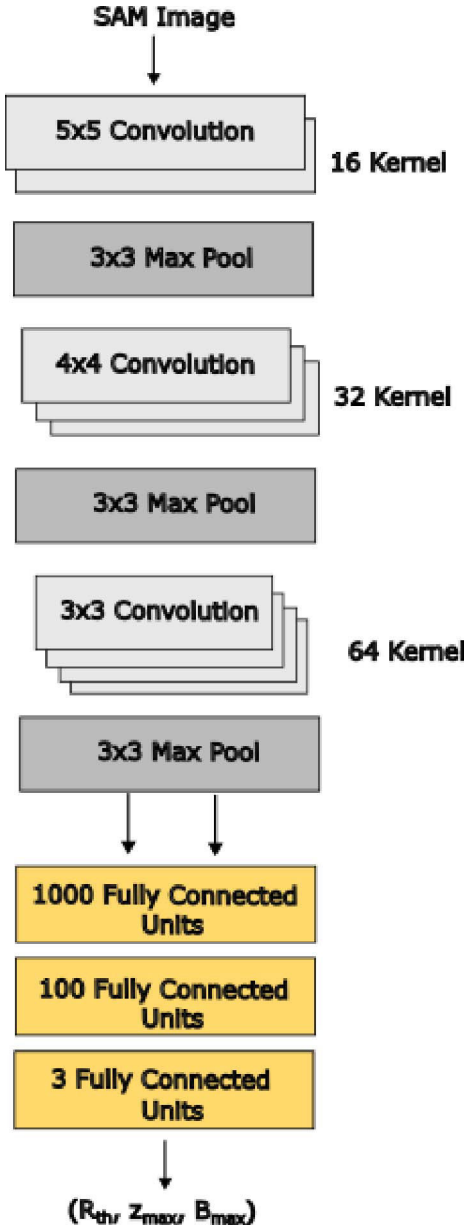


Figure 2: Sketch of model architecture with inputs and outputs. The convolutional layer and pooling layer first find and extract useful features and reduce the input size. This information is then given to the fully connected layers to compute the final outputs.

Care has to be taken in training the model. Since the number of free parameters is large, a model can learn results 'by heart', including noise, instead of capturing the underlying relationships. Such a model will not generalize to new data and is therefore useless. To avoid this, the evaluation of the model is always done on a separate dataset that was not used in training.

B. State of the Art

Reference [14] used a 1D CNN for classification based on features extracted from raw SAM data of defective and intact solder bumps of for two case studies. A similar approach was followed by [15]to classify the porosity of additive manufacturing samples based on raw SAM data using a 1D CNN model as well as a model built purely from fully connected layers and one model using multi-layer

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perceptrons. Here, the CNN performed best, though all models were decently accurate.

An architecture similar to ours (see Figure 2 and the next section) was used in [16] to classify SAM images of a power device into normal and abnormal for monitoring based on a very limited dataset.

IV. Neural Network implementation

We used the architecture shown in Figure 2. The input is the 2D SAM image of a solder joint for one of the four cycles and the outputs are the corresponding z_{max}, B_{max} and R_{th} of the LED for this cycle. Tensorflow was used for the implementation of the architecture and the training of the models.

We apply a series of three Convolutional and Max Poling layer pairs followed by three Fully Connected layers to the input images to compute the final output. We decrease the kernel size for each layer while increasing the number of kernels. Initially, we want the kernel to have a wide enough region that important effects are not missed. As the pooling layers are applied, the image shrinks and each pixel contains information about a larger region, so a smaller kernel is sufficient and makes training easier. The increase in the number of kernels is because later layers in a CNN architecture tend towards more complicated features, of which there are more possible types.

The activation function for all layers is the rectifier function $\sigma(x) = \max(0, x)$, which is a very common activation function as it's simple and does not impose a maximum value on the output which can lead to problems in training [17].

We use the default initialization for the trainable parameter; a uniform Glorot distribution [18] for the kernel of the convolutional network and weights of the fully connected layer, while the biases start as zeroes.

The three outputs (z_{max}, B_{max} and R_{th}) have very different value ranges. For the error function used here (mean squared error), this would mean that one has much greater weight and the optimization would focus on that output. To avoid this and consider all outputs with equal importance, the outputs are rescaled with a standard scaler. The mean μ is subtracted and then the value is divided by the standard deviation σ, $x_{scaled} = (x - \mu)/\sigma$. This produces a scaled parameter where most values are in a range of -1 to 1. This also speeds up training, since the distribution of parameters at initialization will produce a model whose outputs are roughly in this value range, and so no training steps are needed to first reach the correct range.

Since the dataset was small compared to the variation in factors (5 solders, 9 LED types, 4 cycles), splitting the data into Train/Test sets had to be done carefully, so that chance does not produce unrepresentative sets, especially when considering subsets that are further reduced in size. We therefore split each solder/component/cycle combination separately to enforce correct proportions of each combination in each set. A 85%/15% split ratio was used.

The model was trained using the ADAM optimizer [19], which requires very little hyperparameter tuning, using the standard Mean Square Error. We used the default Tensorflow settings for ADAM (learning rate 0.001, β_1=0.9, β_1=0.999, ϵ=0.9).

Training was for 500 epochs, with an early stop if no improvement over a period of 100 epochs was made. This was mostly to speed up training, since overfitting was not a significant issue even at 500 epochs.

We trained several models on different subsets of the complete dataset to understand how the data available during training determines the ability of the network to apply to other data sets.

V. Results

In this section, we discuss the results of our trained models. We focus on the inter- and extrapolation ability of the trained models along three different axes: Between solders, between different LED types, and between different cycles.

We use confusion matrices to display the results. The rows are the different models that result from training on the dataset of the row, while the columns show the dataset the model was evaluated on. Finally, the cell shows the performance of the specific model on the specific dataset in terms of mean square error.

'all' is the full test dataset. 'only X' means only measurements of type X were used, i.e. 'only SAC105' means only LEDs soldered with SAC105. 'no X' means that the full dataset except for X was used, i.e. 'no 500 cycles' means that all measurements except those taken at 500 cycles.

For example, the second cell in the first row of Figure 3 shows a mean square error of 0.16. This means that the model trained on the full dataset ('all') evaluated on the LEDs soldered with SAC105 ('only SAC105') had on average a square error of 0.16.

A. Solders

The results for the solders are shown in Figure 3.

To start with, it's clear that the solder has an effect, which is to be expected since it impacts the aging behavior, and the resulting cracks impact the heat flow and therefore the thermal impedance. There is also an influence on the initial void ratio, which impacts the thermal impedance for the same reason. However, the difference due to the solder is quite small, see the discussion in [2], so this is a very minor effect. The model architecture can handle differences in solder, especially once it has seen examples of a solder.

Second, the cross diagonal has the best performance. This is natural, since this corresponds to testing on the type of data it was trained on. The effect is quite strong in the upper left quadrant, where the model was trained on a small, specialized dataset, and faint but still existent on the lower right quadrant, where the dataset was much larger and varied. The inverse of this effect can be seen in the lower left quadrant, where the diagonal is worse. This also makes sense, since this means testing on a completely new type; requiring extrapolation. This effect is also seen in the top right quarter, but it is quite weak since the datasets there are more similar to each other. The effect of the individual models is much stronger.

Different solders are also more or less well suited to extrapolation. SAC327+Sb is particularly easy to extrapolate to (see the 'only SAC327+Sb' column) but quite difficult to extrapolate from (see 'only SAC327+Sb' row). SAC387+SbBiNi is similar, though less extreme. The cause is likely the higher crack resistance of these two solders, which have the highest silver fraction and are the most reliable. The dataset contains much fewer examples of highly

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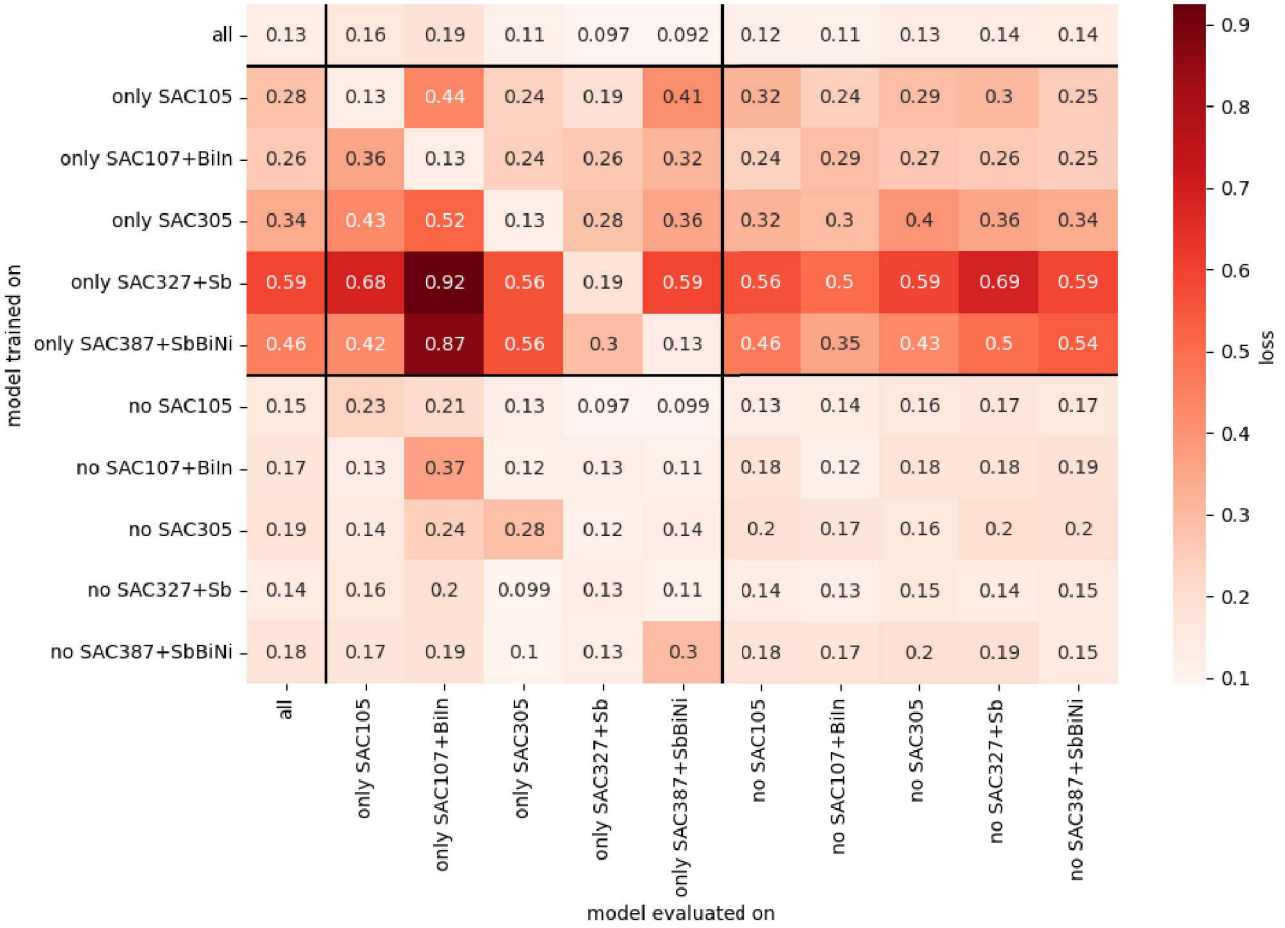


Figure 3: Inter- and extrapolation for solder modells. Shown is the confusion matrix of mean squared error for the dataset the model is trained on (rows) vs the dataset the model is evaluated on (columns). The lines separate different categories of dataset (full dataset, only one type of solder, all except for one type of solder).

damaged joints, which means a model trained on this dataset will be confronted with unfamiliar examples and perform worse when dealing with other solders. The inverse effect can be seen with the two low silver solders SAC105 and SAC107+BiIn, though SAC107+BiIn shows stronger effect than SAC105, despite the fact that it should be the more reliable solder and therefore with a weaker effect.

B. LED types

Figure 4 shows the confusion matrix for the LED types. Note that the error is logarithmic, because the FC-SP1 and VTF1 based models perform worse by several orders of magnitude. We will discuss these first, and then look at the rest based on Figure 5, where these two LED types are removed and the loss is shown normally.

The bad performance of the model based on these two LEDs is because both are of a different type compared to the rest (copper submount vs ceramic submount) which strongly reduces the thermal stress during aging and therefore the crack propagation. We again have the problem that highly cracked states are not reached. As another problem, these different submounts also have different thermal properties. Finally, VTF1 is much larger with a body size of 14 mm^2 compared to 2-5 mm^2 of the other LEDs. Still, the models based on these two are notably bad, even when trying to predict the dataset they were trained for. They do much better there, but still worse than all the other models predicting their own dataset. They also cannot predict each other, so it's not solely an issue for ceramic vs copper submount. One possible reason why extrapolating from one to the other does not work is that these two also have by far the smallest (FC-SP1) and largest (VTF1) pad areas in the study.

On the other hand, FC-SP1 and VTF1 clearly can be predicted so long as the model was trained with examples, as seen in the 'only VTF1' and 'only FC-SP1' columns, particularly in the lower left quadrant. VTF1 is in fact by far the easiest to predict overall in this sector, and also for the model trained on the full dataset. However, it's no full extrapolation, as the 'no VTF1' model performs rather badly. While data from VTF1 LEDs is required for good performance, extra outside data moves the performance from quite bad (compare the values in the upper left quadrant diagonal) to notably good (the models in the lower left quadrant have the best performance overall on the only VTF1 dataset). This is similar to the very reliable the high silver solders, as the carrier submounts show very little degradation over aging.

For the remaining LEDs, FC-GB3 stands out as particularly notable, see Figure 5. Predicting the behaviour of this LED is quite difficult. One possible cause is the thinfilm metallization between the carrier and the solder of the FC-GB2 and FC-GB3 LEDs compared to other LED types. This is substantiated by the fact that model trained on the FC-GB2 does better than any other that does not contain FC-GB3

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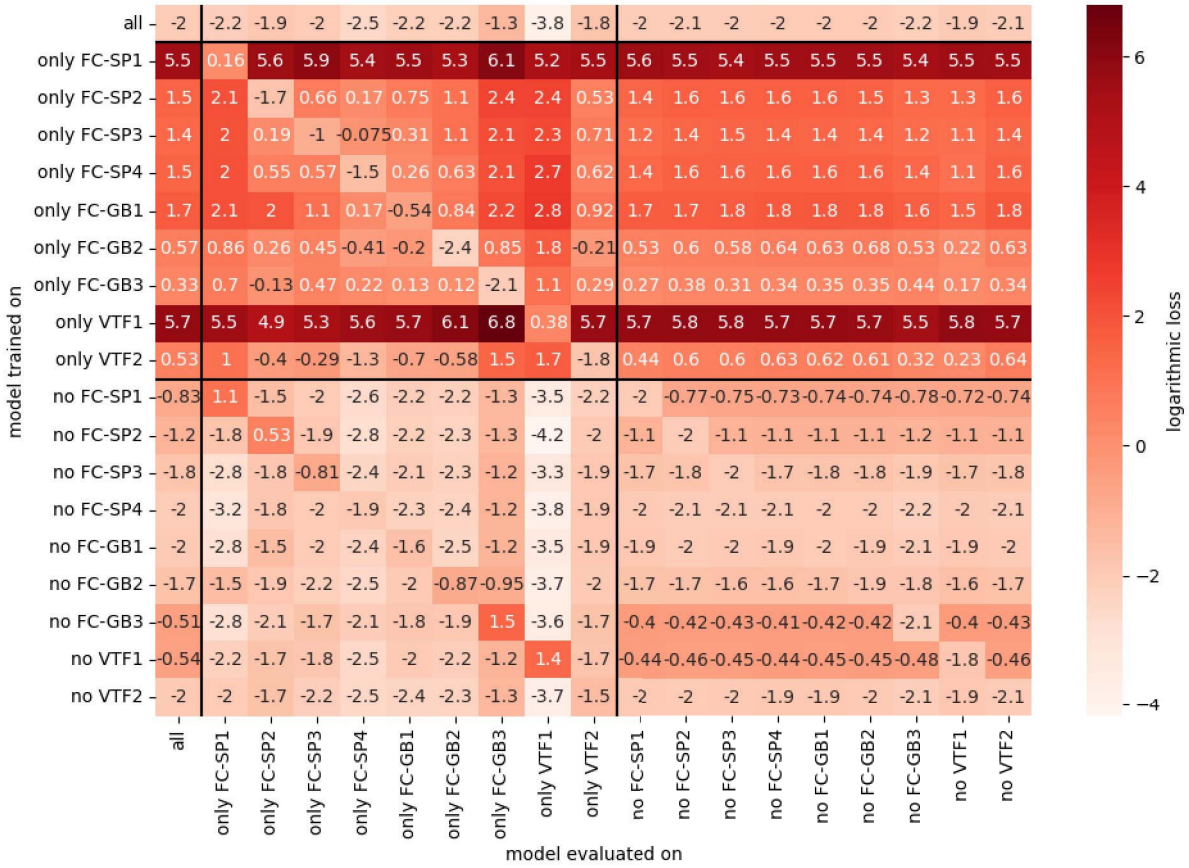


Figure 4: Inter- and extrapolation for LED type models. Shown is the confusion matrix of logarithmic mean squared error for the dataset the model is trained on (rows) vs the dataset the model is evaluated on (columns). The log-error is chosen because the VTF1 and FC-SP1 models perform orders of magnitude worse than the other models, and make display much more difficult. The lines separate different categories of dataset (full dataset, only one type of LED, all except for one type of LED).

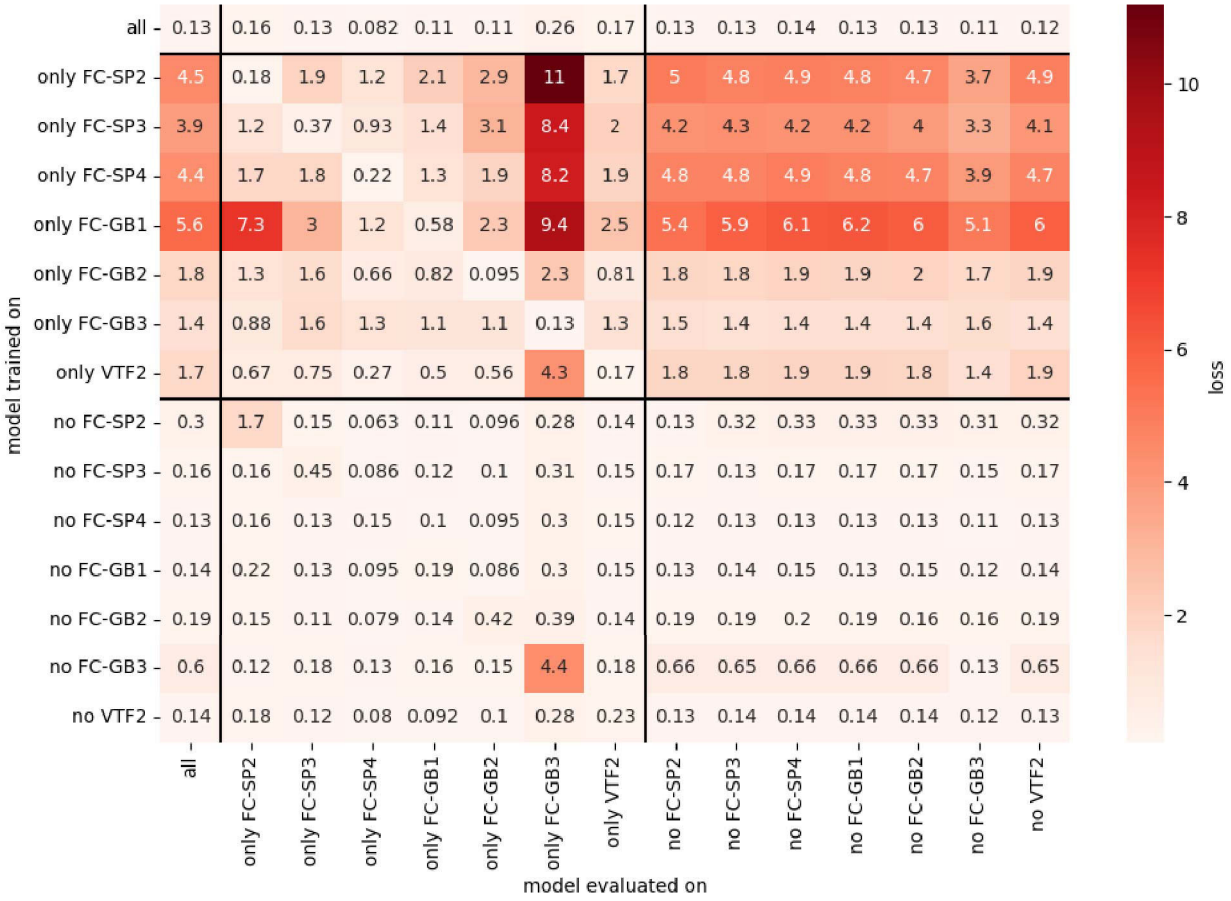


Figure 5: Inter- and extrapolation for LED type modells. Shown is the confusion matrix of mean squared error for the dataset the model is trained on (rows) vs the dataset the model is evaluated on (columns). VTF1 and FC-SP1 datasets are not shown. The lines separate different categories of dataset (full dataset, only one type of LED, all except for one type of LED).

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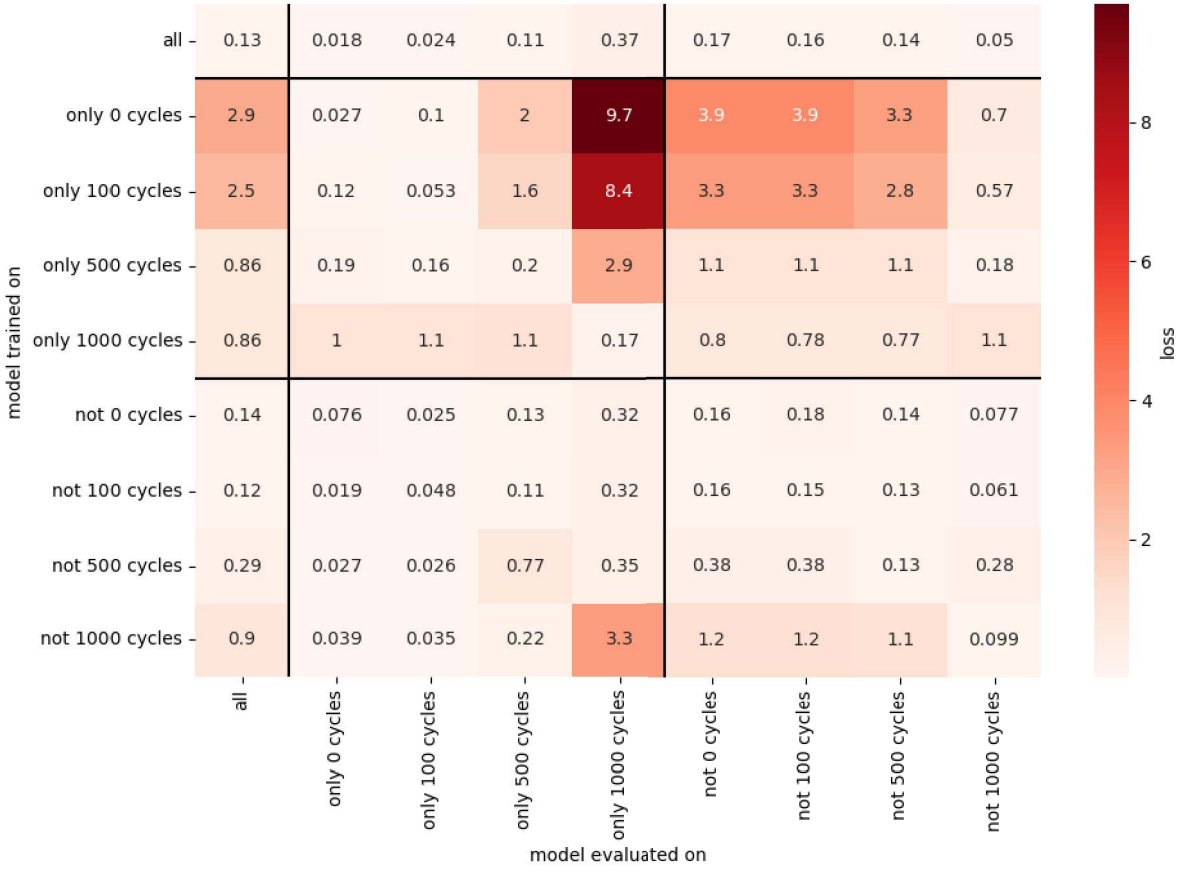


Figure 6: Inter- and extrapolation for cycle models. Shown is the confusion matrix of mean squared error for the dataset the model is trained on (rows) vs the dataset the model is evaluated on (columns). The lines separate different categories of dataset (full dataset, only one cycle, all except for one cycle).

LEDs. Looking at the 'no FC-GB3' model shows that this effect can be diluted, since this model is quite a bit worse than the one trained on only the FC-GB2 LEDs for translating FC-GB3. This is despite the fact that the 'no FC-GB3' model contains the FC-GB2 as a subset. This is a case where more data actually degraded performance.

The upper right quadrant and the all column show that FC-GB2, FC-GB3 and VTF2 extrapolate much better to the full data set than the other types. This might be because these LEDs age quicker, similar to how it the effect for the solders.

C. Cycles

Figure 6 shows the confusion matrix for training based on different cycles. We see that the 0 and 100 cycle models are rather similar. This is because not much has happened at this point since crack growth has not really started. Both are unable to really predict the heavily damaged state at 1000 cycles and take a significant performance hit for 500 cycles.

The 500 cycles model can extrapolate back to the early cycles almost as well as the models trained on those, and handles the 1000 cycles much better, though performance is still far from great. The model for 1000 cycles loses the ability to predict the earlier cycles well, but can handle the later state at 500 cycles. On the overall dataset it is matched with the 500-cycle model.

The data for 0 and 100 cycles is essentially redundant, as can be seen in the lower right quadrant. Missing either does not significantly impact performance. Missing 500 produces a small but notable hit, while missing 1000 is by far the most significant. It is required for decent performance on 1000 cycle data. Extrapolation forward is difficult, but interpolation backwards is much easier.

Figure 7 shows the performance when the data up to a cycle is used to translate images at different cycles. Earlier cycles are easier. The improvement for the dataset including 1000 cycles is especially notable, as it improves on all cycles, including 0 and 100.

There is one outlier with the 0+100+500 cycle model used for the 0 cycle images. However, it's likely that this is simply because of the stochastic training performing slightly worse here. Studies on the variability of the model training by repeatedly training the same model show a standard deviation on the order of 0.05 for models evaluated at the 0-cycle error.

Training on both 0 and 100 cycles does improve prediction for the later cycles. This improvement is small, but based on some tests statistically significant. Add 500 cycles leads to a very big improvement on 1000 cycles, though the performance is still fairly bad. This indicates that the partially damaged solder pads are now handled, but the heavily damaged pads still have big errors.

VI. Conclusions and Outlook

We showed that machine learning can be used to translate SAM measurements to TTA values. We further considered how generalizable this process is.

It's clear that unprecedented physical features can significantly reduce accuracy, as is to be expected. A more

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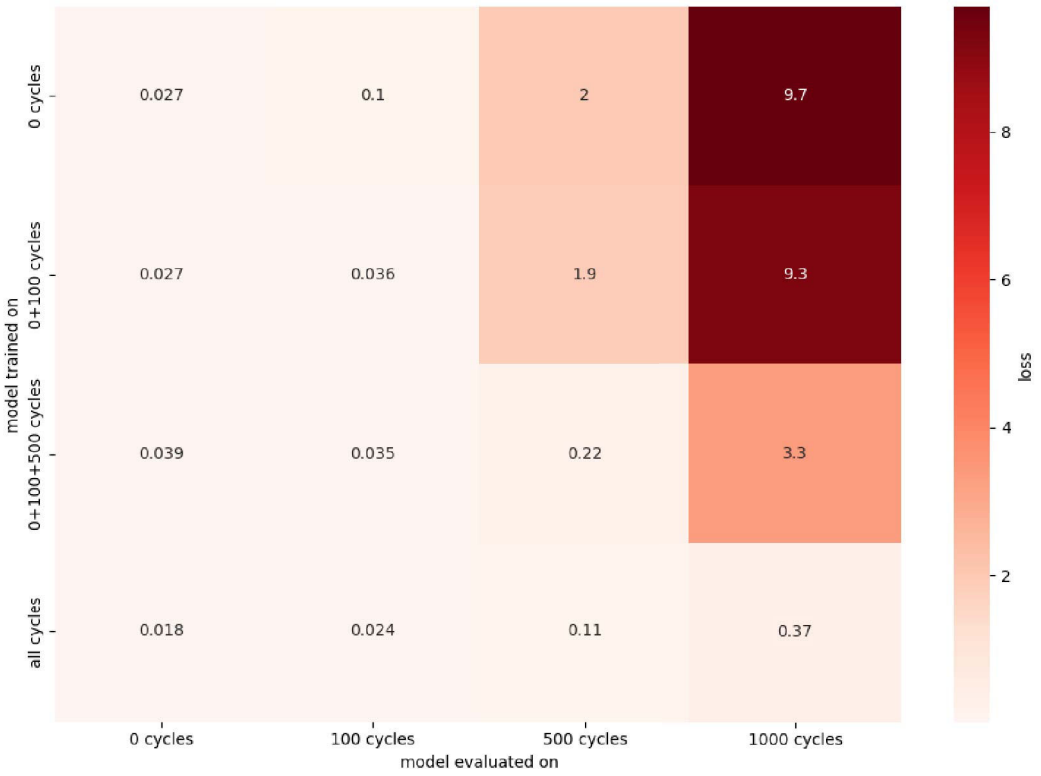


Figure 7: Inter- and extrapolation based on cumulative cycles, i.e. all measurements up to a cycle. Shown is the confusion matrix of the mean squared error for the dataset the model is trained on (rows) vs the dataset the model is evaluated on (columns).

varied dataset could reduce this, but the number of parameters that can strongly impact things is quite large. However, for small variations, interpolation can work quite well. This can be seen for the solders. A possible extension to improve performance would be to give some material properties of the solder to the network in addition to the SAM image to improve performance on new data.

The next step important for this approach is to reverse the direction, and attempt to make statements about the SAM measurement based on TTA data. This would allow statements about the quite interesting but laborious and slow to measure SAM with the much quicker and easier TTA data. Prior work has shown that the TTA data does contain some spatial information [7] [8], so statements about for example the crack ratio of the different solder pads are possible.

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